Basics of Probability and Statistics

Chapter 5 – Group Presentation

Department of Computer Science and Engineering
National Institute of Technology Hamirpur

Presented by:
Group – 1
22dcs001 to 22dcs007

Presented to:
Dr. Pooja Sharma

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Introduction to Probability & Statistics

Why Probability Is Important in ML

- Real-world data is inherently **noisy** and **incomplete**.
- ML models must *quantify uncertainty* to make reliable predictions.
- Probability provides a principled framework for:
  - Reasoning under uncertainty
  - Making optimal decisions
  - Learning from data
- Without probability, we cannot distinguish confident predictions from random guesses.

Role of Uncertainty in Data

Sources of Uncertainty

1. **Aleatoric uncertainty** – inherent randomness in data (e.g. sensor noise).
2. **Epistemic uncertainty** – lack of knowledge, reducible with more data.

- Uncertainty affects every stage: data collection, feature extraction, prediction.
- Probabilistic models can *separate* these sources and handle each appropriately.
- Crucial in safety-critical systems (medical, autonomous vehicles).

Deterministic vs. Probabilistic Models

Deterministic

- Fixed output for a given input
- $y = f(x) = Wx + b$
- No notion of confidence
- Simpler but less informative

Probabilistic

- Output is a *distribution*
- $P(y \mid x; \theta)$
- Captures prediction uncertainty
- Enables risk-aware decisions

Most modern ML models are inherently probabilistic.

Examples in ML

1. **Classification probability:**
   - Softmax outputs give $P(\text{class} \mid \text{input})$.
   - Example: "90% cat, 8% dog, 2% bird".
2. **Prediction uncertainty:**
   - Regression models can output $\mu \pm \sigma$ (mean $\pm$ std. dev.).
   - Bayesian Neural Networks provide full posterior distributions.
3. **Anomaly detection:**
   - Data points with low probability under the learned distribution are flagged.

Real-World Examples

Spam Detection

- Naive Bayes classifier uses $P(\text{spam} \mid \text{words})$.
- Computes likelihood of each word appearing in spam vs. ham.
- Decision threshold balances false positives/negatives.

Medical Diagnosis

- $P(\text{disease} \mid \text{symptoms, tests})$.
- Bayes' theorem updates diagnosis as new test results arrive.
- Probability helps doctors weigh risks vs. benefits of treatment.

Numerical Example: Softmax Classification

Problem

A neural network outputs logits $\mathbf{z} = [2.0,\ 1.0,\ 0.5]$ for classes Cat, Dog, Bird. Compute the softmax probabilities.

Solution:

$$\hat{y}_c = \frac{e^{z_c}}{\sum_j e^{z_j}}$$

$$e^{2.0} = 7.389, \quad e^{1.0} = 2.718, \quad e^{0.5} = 1.649$$

$$\text{Sum} = 7.389 + 2.718 + 1.649 = 11.756$$

$$P(\text{Cat}) = \frac{7.389}{11.756} = \mathbf{0.629}, \quad P(\text{Dog}) = \frac{2.718}{11.756} = \mathbf{0.231}$$

$$P(\text{Bird}) = \frac{1.649}{11.756} = \mathbf{0.140}$$

Prediction: Cat (62.9% confidence). ✓ $\sum P = 1.0$

Fundamentals of Probability

Sample Space and Events

- A **random experiment** is a process whose outcome is uncertain.
- The **sample space** Ω is the set of all possible outcomes.
- An **event** A is a subset of the sample space: $A \subseteq \Omega$.

Examples

- Coin toss: $\Omega = \{H, T\}$; event $A = \{H\}$.
- Rolling a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$; event "even" $= \{2, 4, 6\}$.
- ML: predicting a class label $\Omega = \{c_1, c_2, \ldots, c_k\}$.

Axioms of Probability

Kolmogorov Axioms:

1. **Non-negativity:** $P(A) \geq 0$ for every event A.
2. **Normalization:** $P(\Omega) = 1$.
3. **Additivity:** For mutually exclusive events $A_1, A_2, \ldots$:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Consequences:

- $P(\emptyset) = 0$
- $P(A^c) = 1 - P(A)$
- $0 \leq P(A) \leq 1$

Conditional Probability

Definition

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

- The probability of event A occurring *given* that B has occurred.
- Fundamental to updating beliefs with new evidence.
- **Example:** $P(\text{spam} \mid \text{“free”})$ – probability an email is spam given it contains the word “free”.

Independence of Events

Two events A and B are **independent** if:

$$P(A \cap B) = P(A) \cdot P(B)$$

Equivalently: $P(A \mid B) = P(A)$ – knowing B tells us nothing about A.

- **Naive Bayes** assumes all features are conditionally independent given the class.
- True independence is rare in practice, but the assumption often yields surprisingly good results.
- **Mutual independence** of n events requires all $2^n - n - 1$ subset conditions to hold.

Law of Total Probability

If $B_1, B_2, \ldots, B_n$ form a **partition** of Ω (mutually exclusive, exhaustive):

$$\boxed{P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i)}$$

- Breaks a complex probability into simpler conditional pieces.
- Essential building block for **Bayes' theorem**.
- **ML application:** Marginalising over latent variables:

$$P(x) = \sum_{z} P(x \mid z)\, P(z)$$

Numerical Example: Conditional Probability

Problem

A bag has 5 red and 3 blue balls. Two balls are drawn without replacement. Find $P(\text{2nd is red} \mid \text{1st is red})$.

Solution:

- After drawing 1 red ball: remaining = 4 red + 3 blue = 7 balls.
- $P(\text{2nd red} \mid \text{1st red}) = \frac{4}{7} \approx \mathbf{0.571}$

Verification using formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{5/8 \times 4/7}{5/8} = \frac{4}{7} \approx 0.571 \checkmark$$

Numerical Example: Law of Total Probability

Problem

Factory has 3 machines. Machine A produces 50% of items (2% defective), Machine B produces 30% (3% defective), Machine C produces 20% (5% defective). Find the probability that a randomly chosen item is defective.

Solution:

$$\begin{aligned} P(D) &= P(D \mid A)\,P(A) + P(D \mid B)\,P(B) + P(D \mid C)\,P(C) \\ &= (0.02)(0.50) + (0.03)(0.30) + (0.05)(0.20) \\ &= 0.010 + 0.009 + 0.010 \\ &= \mathbf{0.029} \quad (2.9\% \text{ defective overall}) \end{aligned}$$

Random Variables

Discrete vs. Continuous Random Variables

Discrete Random Variable

- Takes countable values
- Examples: class labels, word counts, pixel categories
- Described by a **PMF**

Continuous Random Variable

- Takes uncountably many values
- Examples: weights, pixel intensities, loss values
- Described by a **PDF**

A random variable $X : \Omega \to \mathbb{R}$ maps outcomes to real numbers.

Probability Mass Function (PMF)

For a discrete random variable X:

$$p(x) = P(X = x)$$

Properties:

1. $p(x) \geq 0$ for all x.
2. $\sum_{\text{all } x} p(x) = 1$.

Example: Fair Die

$p(x) = \frac{1}{6}$ for $x \in \{1, 2, 3, 4, 5, 6\}$; $\quad p(x) = 0$ otherwise.

ML usage: Softmax output is a PMF over class labels.

Probability Density Function (PDF)

For a continuous random variable X:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Properties:

1. $f(x) \geq 0$ for all x.
2. $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$.
3. Note: $P(X = x) = 0$ for any single point.

Example: Gaussian PDF

$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

Cumulative Distribution Function (CDF)

Definition

$$F(x) = P(X \leq x) = \begin{cases} \sum_{t \leq x} p(t) & \text{(discrete)} \\ \int_{-\infty}^{x} f(t)\,dt & \text{(continuous)} \end{cases}$$

Properties:

- F is non-decreasing: $x_1 < x_2 \Rightarrow F(x_1) \leq F(x_2)$.
- $\lim_{x \to -\infty} F(x) = 0, \quad \lim_{x \to \infty} F(x) = 1$.
- $P(a < X \leq b) = F(b) - F(a)$.

ML usage: Quantile functions, threshold selection for classifiers.

Expected Value (Mean of a Random Variable)

Definition

$$\mathbb{E}[X] = \mu = \begin{cases} \sum_x x\, p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x\, f(x)\, dx & \text{(continuous)} \end{cases}$$

Key Properties:

- Linearity: $\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$.
- $\mathbb{E}[g(X)] = \sum_x g(x)\, p(x)$ (Law of the Unconscious Statistician).

ML usage:

- Loss functions are *expected* losses over the data distribution.
- SGD estimates $\mathbb{E}[\nabla \mathcal{L}]$ using mini-batches.

Numerical Example: PMF & Expected Value

Problem

A classifier predicts classes $\{0, 1, 2\}$ with PMF: $p(0) = 0.3$, $p(1) = 0.5$, $p(2) = 0.2$. Find $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

Solution:

$$\begin{aligned}\mathbb{E}[X] &= \sum_x x\,p(x) = 0(0.3) + 1(0.5) + 2(0.2) \\ &= 0 + 0.5 + 0.4 = \mathbf{0.9}\end{aligned}$$

$$\begin{aligned}\mathbb{E}[X^2] &= 0^2(0.3) + 1^2(0.5) + 2^2(0.2) \\ &= 0 + 0.5 + 0.8 = \mathbf{1.3}\end{aligned}$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 1.3 - 0.81 = \mathbf{0.49}$$

Numerical Example: CDF Computation

Problem

Given the PMF: $p(1) = 0.2$, $p(2) = 0.3$, $p(3) = 0.35$, $p(4) = 0.15$.
Find the CDF $F(x)$ and compute $P(2 < X \leq 4)$.

Solution:

x	1	2	3	4
$p(x)$	0.20	0.30	0.35	0.15
$F(x)$	0.20	0.50	0.85	1.00

$$P(2 < X \leq 4) = F(4) - F(2) = 1.00 - 0.50 = \mathbf{0.50}$$

Probability Distributions Used

Bernoulli Distribution

Models a single binary outcome (success/failure).

$$X \sim \text{Bernoulli}(p), \quad P(X = 1) = p, \;\; P(X = 0) = 1 - p$$

- $\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$
- **ML usage:** Binary classification (spam/not-spam), logistic regression output.
- Sigmoid function maps logits to $p \in (0, 1)$.

Binomial Distribution

Sum of n independent Bernoulli trials.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

- $\mathbb{E}[X] = np, \quad \mathrm{Var}(X) = np(1-p).$
- **ML usage:** Number of correct predictions in n samples.
- Approaches Gaussian for large n (Central Limit Theorem).

Gaussian (Normal) Distribution

PDF

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Parameterised by mean μ and variance σ^2.
- Symmetric, bell-shaped curve.
- **68–95–99.7 rule:** $\approx$68% within $\pm 1\sigma$, 95% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.
- The most important distribution in statistics and ML.

Poisson & Uniform Distributions

Poisson Distribution

$$P(X = k) = \frac{\lambda^k \, e^{-\lambda}}{k!}$$

- Models count of rare events
- $\mathbb{E}[X] = \mathrm{Var}(X) = \lambda$
- ML: event counting, arrival rates

Uniform Distribution

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

- All values equally likely
- $\mathbb{E}[X] = \frac{a+b}{2}$
- ML: random initialisation, non-informative prior

Applications of Distributions in ML

1. **Noise Modelling:**
   - Sensor/measurement noise $\sim \mathcal{N}(0, \sigma^2)$.
   - Assuming Gaussian noise leads to MSE loss.
2. **Feature Distributions:**
   - Understanding data distribution guides preprocessing choices.
   - Gaussian Mixture Models (GMMs) for clustering.
3. **Bayesian Models:**
   - Prior distributions encode domain knowledge.
   - Posterior distributions capture updated beliefs after observing data.
   - Gaussian priors $\Leftrightarrow$ L2 regularisation.

Numerical Example: Binomial Distribution

Problem

A model has 80% accuracy ($p = 0.8$). In a batch of $n = 5$ predictions, find $P(\text{exactly 4 correct})$ and $P(\text{at least 4 correct})$.

Solution:

$$P(X = 4) = \binom{5}{4}(0.8)^4(0.2)^1 = 5 \times 0.4096 \times 0.2 = \mathbf{0.4096}$$

$$P(X = 5) = \binom{5}{5}(0.8)^5(0.2)^0 = 1 \times 0.3277 \times 1 = \mathbf{0.3277}$$

$$P(X \geq 4) = P(X = 4) + P(X = 5) = 0.4096 + 0.3277 = \mathbf{0.7373}$$

$\Rightarrow$ There is a 73.7% chance of getting 4 or more correct in the batch.

Numerical Example: Gaussian Distribution

Problem

Feature values follow $X \sim \mathcal{N}(\mu = 50,\, \sigma^2 = 25)$, so $\sigma = 5$.
Find $P(45 \leq X \leq 60)$.

Solution: Convert to standard normal $Z = \frac{X-\mu}{\sigma}$:

$$Z_1 = \frac{45-50}{5} = -1.0, \qquad Z_2 = \frac{60-50}{5} = 2.0$$

$$\begin{aligned} P(45 \leq X \leq 60) &= P(-1.0 \leq Z \leq 2.0) \\ &= \Phi(2.0) - \Phi(-1.0) \\ &= 0.9772 - 0.1587 = \mathbf{0.8185} \end{aligned}$$

$\Rightarrow$ About 81.9% of feature values fall between 45 and 60.

Statistical Measures in Data

Mean, Median, Mode

Mean $\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$ (arithmetic average)

Median Middle value of sorted data (robust to outliers)

Mode Most frequently occurring value

- In a symmetric distribution: mean $\approx$ median $\approx$ mode.
- Skewed data: median is preferred over mean.
- **ML:** Batch mean used in Batch Normalisation; median for robust estimation.

Variance & Standard Deviation

Formulas

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2, \qquad \sigma = \sqrt{\sigma^2}$$

- **Variance** σ^2: average squared deviation from the mean.
- **Standard deviation** σ: same units as the data.
- High variance $\Rightarrow$ data is spread out.
- Low variance $\Rightarrow$ data is clustered near the mean.
- **ML:** Used in feature scaling, Batch Normalisation, model evaluation.

Covariance & Correlation

Covariance:

$$\mathrm{Cov}(X, Y) = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})(y_i - \bar{y})$$

Pearson Correlation Coefficient:

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \, \sigma_Y}, \quad -1 \leq \rho \leq 1$$

- $\rho = 1$: perfect positive correlation; $\rho = -1$: perfect negative; $\rho = 0$: no linear correlation.
- **ML:** Covariance matrix is central to PCA; correlation helps detect redundant features.

Importance in ML

1. **Feature Scaling:**
   - Prevents features with large ranges from dominating gradient updates.
2. **Data Normalisation:**
   - **Min-Max:** $x' = \frac{x - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}$
   - **Z-Score:** $x' = \frac{x - \mu}{\sigma}$ (zero mean, unit variance)
3. **Understanding Spread of Data:**
   - Helps identify outliers and class imbalance.
   - Guides choice of model and loss function.

Numerical Example: Variance & Standard Deviation

Problem

Dataset: $X = \{4, 8, 6, 5, 7\}$. Compute the mean, variance, and standard deviation.

Solution:

$$\mu = \frac{4+8+6+5+7}{5} = \frac{30}{5} = \mathbf{6.0}$$

$$\sigma^2 = \frac{(4-6)^2 + (8-6)^2 + (6-6)^2 + (5-6)^2 + (7-6)^2}{5}$$

$$= \frac{4+4+0+1+1}{5} = \frac{10}{5} = \mathbf{2.0}$$

$$\sigma = \sqrt{2.0} = \mathbf{1.414}$$

Numerical Example: Covariance, Correlation & Z-Score

Problem

$X = \{1, 2, 3, 4, 5\}$, $Y = \{2, 4, 5, 4, 5\}$. Find $\mathrm{Cov}(X, Y)$ and ρ_{XY}.
Also, apply Z-score normalisation to X.

Solution: $\bar{x} = 3$, $\bar{y} = 4$, $\sigma_X = \sqrt{2} \approx 1.414$, $\sigma_Y = \sqrt{1.2} \approx 1.095$.

$$\mathrm{Cov}(X, Y) = \frac{(1-3)(2-4) + (2-3)(4-4) + (3-3)(5-4) + (4-3)(4-4) + (5-3)(5-4)}{5}$$

$$= \frac{4 + 0 + 0 + 0 + 2}{5} = \mathbf{1.2}$$

$$\rho_{XY} = \frac{1.2}{1.414 \times 1.095} = \frac{1.2}{1.549} = \mathbf{0.775}$$

Z-Score of X: $x_i' = \frac{x_i - 3}{1.414} \Rightarrow \{-1.41,\ -0.71,\ 0,\ 0.71,\ 1.41\}$

Bayes' Theorem and Bayesian Thinking

Bayes' Theorem

Formula

$$\boxed{P(A \mid B) = \frac{P(B \mid A)\, P(A)}{P(B)}}$$

Prior $P(A)$ – our initial belief before seeing data.

Likelihood $P(B \mid A)$ – how likely the data is under hypothesis A.

Posterior $P(A \mid B)$ – updated belief after seeing data.

Evidence $P(B)$ – normalising constant (total probability of data).

Bayes' theorem is the engine that turns data into knowledge.

Prior, Likelihood, and Posterior

- **Prior** $P(\theta)$**:** Encodes domain knowledge before observing data.
  - Informative prior: strong belief (e.g. expert knowledge).
  - Non-informative prior: uniform or vague (lets data speak).
- **Likelihood** $P(\mathcal{D} \mid \theta)$**:** How well parameters θ explain the data $\mathcal{D}$.
- **Posterior** $P(\theta \mid \mathcal{D})$**:** Combines prior and likelihood.

$$P(\theta \mid \mathcal{D}) \propto P(\mathcal{D} \mid \theta)\, P(\theta)$$

- As more data arrives, the posterior becomes dominated by the likelihood.

Naive Bayes Classifier

Classifies by computing:

$$\hat{c} = \arg\max_{c} \; P(c) \prod_{i=1}^{d} P(x_i \mid c)$$

- Assumes **conditional independence** of features given the class.
- Despite this "naive" assumption, works remarkably well for:
  - Text classification and spam filtering
  - Sentiment analysis
  - Medical diagnosis screening
- Very fast to train – only requires counting frequencies.
- Variants: Gaussian NB, Multinomial NB, Bernoulli NB.

Bayesian Inference

- Instead of finding a single best θ, maintain a **distribution** over θ.
- **Prediction:** Marginalise over all possible parameters:

$$P(y^* \mid x^*, \mathcal{D}) = \int P(y^* \mid x^*, \theta)\, P(\theta \mid \mathcal{D})\, d\theta$$

- Naturally handles model uncertainty and prevents overfitting.
- **Challenges:** Often computationally intractable – use approximations:
  - Variational Inference (VI)
  - Markov Chain Monte Carlo (MCMC)
  - Monte Carlo Dropout (practical approximation)

Applications of Bayesian Thinking

Spam Filtering

- Update spam probability as new emails arrive
- Adapt to evolving spam patterns

Medical Diagnosis

- Combine test results with base rates
- Avoid false-positive pitfalls

Text Classification

- Topic modelling (LDA)
- Bayesian sentiment analysis

Key Insight

Bayesian methods let us incorporate prior knowledge and **update beliefs** as evidence accumulates – a natural framework for learning.

Numerical Example: Bayes' Theorem (Medical Test)

Problem

A disease affects 1% of the population. A test has 95% sensitivity (true positive rate) and 90% specificity (true negative rate).
If a person tests positive, what is $P(\text{disease} \mid \text{positive})$?

Solution:

$$P(D) = 0.01, \quad P(+ \mid D) = 0.95, \quad P(+ \mid D^c) = 0.10$$

$$\begin{aligned} P(+) &= P(+ \mid D)\, P(D) + P(+ \mid D^c)\, P(D^c) \\ &= (0.95)(0.01) + (0.10)(0.99) = 0.0095 + 0.099 = 0.1085 \end{aligned}$$

$$P(D \mid +) = \frac{P(+ \mid D)\, P(D)}{P(+)} = \frac{0.0095}{0.1085} = \mathbf{0.0876} \approx \mathbf{8.8}\%$$

Insight: Despite a "positive" test, there's only 8.8% chance of disease!

Numerical Example: Naive Bayes Classifier

Problem

Classify email with words "free" and "offer" as Spam or Ham.
Given: $P(S) = 0.4$, $P(H) = 0.6$, $P(\text{free} \mid S) = 0.8$, $P(\text{free} \mid H) = 0.1$, $P(\text{offer} \mid S) = 0.7$, $P(\text{offer} \mid H) = 0.2$.

Solution (assuming feature independence):

$$P(S)\,P(\text{free} \mid S)\,P(\text{offer} \mid S) = 0.4 \times 0.8 \times 0.7 = \mathbf{0.224}$$
$$P(H)\,P(\text{free} \mid H)\,P(\text{offer} \mid H) = 0.6 \times 0.1 \times 0.2 = \mathbf{0.012}$$

$$P(S \mid \text{free, offer}) = \frac{0.224}{0.224 + 0.012} = \frac{0.224}{0.236} = \mathbf{0.949}$$

Prediction: Spam with 94.9% probability.

Statistical Concepts Used

Maximum Likelihood Estimation (MLE)

Given data $\mathcal{D} = \{x_1, \ldots, x_N\}$ and parameters θ:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \prod_{i=1}^{N} P(x_i \mid \theta) = \arg\max_{\theta} \sum_{i=1}^{N} \log P(x_i \mid \theta)$$

- Maximises the probability of observed data under the model.
- Equivalent to minimising the **negative log-likelihood** (NLL).
- Foundation of most supervised learning objectives.
- **Example:** MLE for Gaussian estimates $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \frac{1}{N} \sum (x_i - \bar{x})^2$.

Maximum A Posteriori (MAP)

Incorporates a prior $P(\theta)$:

$$\hat{\theta}_{\text{MAP}} = \arg\max_{\theta} \; P(\mathcal{D} \mid \theta)\, P(\theta) = \arg\max_{\theta} \left[\sum_{i=1}^{N} \log P(x_i \mid \theta) + \log P(\theta) \right]$$

- Gaussian prior on $\theta \Rightarrow$ **L2 regularisation** (weight decay).
- Laplace prior on $\theta \Rightarrow$ **L1 regularisation** (sparsity).
- MAP = MLE when prior is uniform.
- Bridges frequentist (MLE) and Bayesian approaches.

Bias vs. Variance

Bias

Error from overly simplistic model assumptions.
High bias $\Rightarrow$ **underfitting**.

Variance

Error from sensitivity to training data fluctuations.
High variance $\Rightarrow$ **overfitting**.

$$\text{Expected Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$$

- **Goal:** Find optimal model complexity that minimises total error.
- Cross-validation helps estimate this trade-off empirically.

Sampling, Estimation & Hypothesis Testing

- **Sampling:** Drawing representative subsets from a population.
  - Mini-batch SGD samples from training data.
  - Bootstrap sampling for confidence intervals.
- **Estimation:** Using samples to infer population parameters.
  - Point estimates (MLE, MAP) vs. interval estimates (confidence intervals).
- **Hypothesis Testing:**
  - H_0: null hypothesis; H_1: alternative hypothesis.
  - p-value: probability of observing data as extreme under H_0.
  - Used to compare model performances (e.g. paired t-test).

Overfitting and Underfitting

- **Overfitting:** Model memorises training data, fails on unseen data (high variance).
  - Remedies: regularisation, dropout, early stopping, more data.
- **Underfitting:** Model is too simple to capture patterns (high bias).
  - Remedies: increase model capacity, add features, train longer.
- **Detection:**
  - Train loss $\ll$ val loss $\Rightarrow$ overfitting.
  - Both losses high $\Rightarrow$ underfitting.

Applications in ML Algorithms

Logistic Regression

- Models $P(y = 1 \mid x)$ via sigmoid.
- Trained by MLE (cross-entropy loss).
- L2 penalty = MAP with Gaussian prior.

Neural Networks

- Loss functions grounded in MLE.
- Dropout $\approx$ Bayesian approximation.
- Batch Norm uses batch statistics.

Model Evaluation:

- Cross-validation: unbiased estimate of generalisation error.
- Statistical tests to compare classifiers (McNemar's test, paired t-test).
- Confidence intervals on metrics (Precision, Recall, F1).

Numerical Example: MLE for Gaussian

Problem

Given data $\mathcal{D} = \{3, 5, 7, 5, 10\}$. Estimate $\hat{\mu}_{\text{MLE}}$ and $\hat{\sigma}^2_{\text{MLE}}$ assuming data follows a Gaussian distribution.

Solution:

$$\hat{\mu}_{\text{MLE}} = \bar{x} = \frac{3+5+7+5+10}{5} = \frac{30}{5} = \mathbf{6.0}$$

$$\begin{aligned}
\hat{\sigma}^2_{\text{MLE}} &= \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{\mu})^2 \\
&= \frac{(3-6)^2 + (5-6)^2 + (7-6)^2 + (5-6)^2 + (10-6)^2}{5} \\
&= \frac{9+1+1+1+16}{5} = \frac{28}{5} = \mathbf{5.6}
\end{aligned}$$

$\hat{\sigma}_{\text{MLE}} = \sqrt{5.6} \approx \mathbf{2.366}$

Numerical Example: Bias-Variance Decomposition

Problem

True value: $y = 10$. Three models are trained on different samples.

Model A predictions: $\{9.8,\ 10.1,\ 10.0\}$ Model B predictions: $\{8.0,\ 12.0,\ 10.0\}$

Solution:

Model A:

$$\bar{\hat{y}} = 9.967$$
$$\text{Bias} = 9.967 - 10 = -0.033$$
$$\text{Bias}^2 = \mathbf{0.001}$$
$$\text{Var} = \mathbf{0.016}$$

$\Rightarrow$ Low bias, low variance

Model B:

$$\bar{\hat{y}} = 10.0$$
$$\text{Bias} = 10.0 - 10 = 0$$
$$\text{Bias}^2 = \mathbf{0.0}$$
$$\text{Var} = \mathbf{2.667}$$

$\Rightarrow$ Zero bias, high variance

Insight: Model A is preferred – lower total error despite tiny bias.

Summary

1. Probability & statistics are the **mathematical foundation** of ML.
2. Core concepts: sample spaces, Bayes' theorem, distributions, expected values.
3. Statistical measures guide data preprocessing and feature engineering.
4. Bayes' theorem and Bayesian thinking enable principled learning from data.
5. MLE and MAP provide the theoretical basis for training ML models.
6. Understanding bias–variance trade-off is key to building models that generalise.
7. Every ML algorithm – from logistic regression to deep learning – relies on statistics.

References

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Thank You!

Questions?